

Roll No.

TCS-731

**B. TECH. (IT)/B. TECH. (CSE-SE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

DIGITAL FORENSICS

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) While cracking passwords for a forensic investigation, an investigator has to use a variety of techniques and tricks in order to identify or break the passwords. In your opinion, what should be necessary order of steps and why to break passwords effectively in a digital forensic investigation ?
(CO1, CO2, CO3)

OR

- (b) Briefly discuss the benefits and challenges of a digital forensic case investigation.
(CO1, CO2, CO3)
2. (a) Cybercrimes can be classified by using various categories such as cybercrimes for individuals, against property, against nation etc. Present a comprehensive categorization of these cybercrimes with appropriate examples.
(CO1, CO2, CO3)

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(2)

OR

- (b) List out *five* case studies for cyber crimes against individuals and their corresponding preventive measures. (CO1, CO2, CO3)

3. (a) Compare virus, worm and trojan horse. (CO1, CO2, CO3)

OR

- (b) Present and discuss the flowchart of a cyber forensics investigation. (CO1, CO2, CO3)

4. (a) Define digital forensics. On what basis do we classify the various fields associated with digital forensics ? Critically analyze the areas and their categorization. (CO4, CO5)

OR

- (b) Explain the phases involved in the digital forensic investigation model with appropriate diagram. (CO4, CO5)

5. (a) The trade secrets and sensitive information from a company were sold to a competing company. Employee X who is working in that company is suspected to have sold the information. Employee X claims that his system does not have any USB port. Explain how this suspicion was raised and how can it be proven that Employee X was associated with cybercrime. (CO4, CO5)

OR

- (b) The manager of a company suspects that the pay bills maintained in the system has been tampered with and some of the associated files have been deleted. How can the files be recovered and how do you prove that the paybills are tampered with ?

(CO4, CO5)